

Normal Probability Test

— Understanding the Normal Probability Q-Q plot, Order Statistics

STAT 24510-30040

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Question of interest: Are independent observations $X_1, \dots, X_n$ from a common normal distribution?

Normal Probability Q-Q plot

Normal Q-Q plot is a graphical method to check the normality assumption of an observed sample of data.

If $x_1, x_2, \dots, x_n$'s are indeed independent observations from a normal distribution $N(\mu, \sigma^2)$, then their normal Q-Q plot should consist of n points roughly forming a straight line, regardless the values of the mean μ and variance σ^2 of the normal distribution that the observations are sampled from.

In the following we explain the theoretical base of the normal Q-Q plot.

What are plotted on a normal Q-Q plot?

First, we walk through the reasoning and approximations of the Q-Q plot.

- Given $X_1, \dots, X_n$, let

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

be the ordered values of $X_1, \dots, X_n$. The $X_{(i)}$'s are called the **order statistics** of $X_1, \dots, X_n$.

Just as $X_1, \dots, X_n$, the order statistics are random variables, before observations.

If X_i 's are *i.i.d.* $\sim N(\mu, \sigma^2)$, then $X_i = \mu + \sigma Z_i$, where Z_i 's are *i.i.d.* $\sim N(0, 1)$. There should be a linear relationship

$$X_{(i)} = \mu + \sigma Z_{(i)}$$

where

$$Z_{(1)} \leq Z_{(2)} \leq \dots \leq Z_{(n)}$$

denote the order statistics of $Z_1, \dots, Z_n$, with Z_i *i.i.d.* $\sim N(0, 1)$.

If we observe $X_1, \dots, X_n$ i.i.d. $\sim N(\mu, \sigma^2)$ and $Z_1, \dots, Z_n$ i.i.d. $\sim N(0, 1)$, then the n pairs of observations of the order statistics

$$(Z_{(i)}, X_{(i)}), \quad i = 1, \dots, n$$

should resemble a straight line, for any values of μ and σ^2 .

- Approximation $Z_{(i)}$ by its mean, the linear relationship becomes an approximation,

$$X_{(i)} \approx \mu + \sigma \mathbb{E}[Z_{(i)}], \quad i = 1, \dots, n.$$

Still, the n pairs

$$(\mathbb{E}[Z_{(i)}], X_{(i)}), \quad i = 1, \dots, n$$

obtained from observations of $X_1, \dots, X_n$ should approximately form a line.

- Denote $\Phi(z)$ as the cumulative probability function of $N(0, 1)$, then we can show (see below) that

$$\mathbb{E}[\Phi(Z_{(i)})] = \frac{i}{n+1}, \quad i = 1, \dots, n. \quad (1)$$

This provides an approximation of $\mathbb{E}[Z_{(i)}]$,

$$\mathbb{E}[Z_{(i)}] \approx \Phi^{-1}(\mathbb{E}[\Phi(Z_{(i)})]) = \Phi^{-1}\left(\frac{i}{n+1}\right), \quad i = 1, \dots, n$$

which can be easily calculated by any software.

- Given n observed points $x_1, \dots, x_n$ and their ordered values $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$,

Normal Q-Q plot is formed by the points

$$\left(\Phi^{-1}\left(\frac{i}{n+1}\right), x_{(i)} \right), \quad i = 1, \dots, n.$$

The points should resemble a straight line, if x_i ' are indeed *i.i.d.* observations from $N(\mu, \sigma^2)$ for some μ, σ^2 .

Derivations

In the following we prove the equation in (1).

Proof.

The approximation of the expectation of $\mathbb{E}[Z_{(i)}] \approx \Phi^{-1}\left(\frac{i}{n+1}\right)$ is based on $\mathbb{E}[\Phi(Z_{(i)})] = \frac{i}{n+1}$.

To prove the equation $\mathbb{E}[\Phi(Z_{(i)})] = \frac{i}{n+1}$ in (1), first we need the following two results.

Result(a)

If random variable Y has cumulative distribution function $F(y) = \mathbb{P}(Y \leq y)$, then

$$F(Y) \sim \text{Uniform}([0, 1])$$

This is because

$$\mathbb{P}(F(Y) \leq u) = \mathbb{P}(Y \leq F^{-1}(u)) = F(F^{-1}(u)) = u, \quad \forall u \in [0, 1]$$

where the inverse function

$$F^{-1}(u) = \inf\{y : F(y) \geq u\}$$

Result(b)

Assume $Y_1, \dots, Y_n$ are *i.i.d.* with cumulative distribution function $F(y) = \mathbb{P}(Y \leq y)$ and density $f(y) = F'(y)$, and $Y_{(1)} \leq Y_{(2)} \leq \dots \leq Y_{(n)}$ are the order statistics of $Y_1, \dots, Y_n$.

Then the cumulative distribution function of the k th order statistic $Y_{(k)}$ has the form

$$\mathbb{P}(Y_{(k)} \leq y) = \sum_{i=k}^n \binom{n}{i} (F(y))^i (1 - F(y))^{n-i} \quad (2)$$

and the k th order statistic $Y_{(k)}$ has density function

$$f_{(k)}(y) = \frac{\partial}{\partial y} \mathbb{P}(Y_{(k)} \leq y) = n f(y) \binom{n-1}{k-1} (F(y))^{k-1} (1 - F(y))^{n-k} \quad (3)$$

Derivation of order statistics distributions in Result (b)

To derive the formula of $\mathbb{P}(Y_{(k)} \leq y)$ in (2), consider the locations of $Y_i, i = 1, \dots, n$. Note that

$$p_y = F(y) = \mathbb{P}(Y_i \leq y), \quad i = 1, \dots, n$$

is the common probability for any Y_i falls to the left of y . By the law of Binomial,

$$\mathbb{P}(\text{k of the } Y_i \leq y) = \binom{n}{k} p_y^k (1 - p_y)^{n-k} = \binom{n}{k} (F(y))^k (1 - F(y))^{n-k}$$

Therefore

$$\mathbb{P}(Y_{(k)} \leq y) = \mathbb{P}(\text{at least k of the } Y_i \leq y) = \sum_{i=k}^n \binom{n}{i} (F(y))^i (1 - F(y))^{n-i}$$

The formula for density functions of order statistics in (3) can be obtained by taking derivative of their cumulative probability functions.

$$f_{(k)}(y) = \frac{\partial}{\partial y} \mathbb{P}(Y_{(k)} \leq y) = \frac{\partial}{\partial y} \left(\sum_{i=k}^n \binom{n}{i} (F(y))^i (1 - F(y))^{n-i} \right) = \sum_{i=k}^n \binom{n}{i} \frac{\partial}{\partial y} ((F(y))^i (1 - F(y))^{n-i})$$

For $i = k, \dots, n$,

$$\frac{\partial}{\partial y} ((F(y))^i (1 - F(y))^{n-i}) = i(F(y))^{i-1} (1 - F(y))^{n-i} f(y) - (n-i)(F(y))^i (1 - F(y))^{n-i-1} f(y)$$

when the exponents are ≥ 0 . Note that for $i = n$, the second term is zero,

$$\frac{\partial}{\partial y} ((F(y))^i (1 - F(y))^{n-i}) = i(F(y))^{i-1} (1 - F(y))^{n-i} f(y) - 0$$

Then the second sum in $f_{(k)}(y)$ stops at $n-1$,

$$\begin{aligned} f_{(k)}(y) &= f(y) \left(\sum_{i=k}^n \binom{n}{i} i(F(y))^{i-1} (1 - F(y))^{n-i} - \sum_{i=k}^{n-1} \binom{n}{i} (n-i)(F(y))^i (1 - F(y))^{(n-1)-i} \right) \\ &= n f(y) \left(\sum_{i=k}^n \frac{(n-1)!}{(i-1)!(n-i)!} (F(y))^{i-1} (1 - F(y))^{n-i} - \sum_{i=k}^{n-1} \frac{(n-1)!}{i!((n-1)-i)!} (F(y))^i (1 - F(y))^{(n-1)-i} \right) \end{aligned}$$

Substitute the index i by $j = i-1$ in the first sum, then label j by i again,

$$\begin{aligned} f_{(k)}(y) &= n f(y) \left(\sum_{j=k-1}^{n-1} \frac{(n-1)!}{j!(n-j-1)!} (F(y))^j (1 - F(y))^{n-j-1} - \sum_{j=k}^{n-1} \frac{(n-1)!}{i!((n-1)-i)!} (F(y))^i (1 - F(y))^{(n-1)-i} \right) \\ &= n f(y) \left(\sum_{i=k-1}^{n-1} \frac{(n-1)!}{i!((n-1)-i)!} (F(y))^i (1 - F(y))^{n-1-i} - \sum_{j=k}^{n-1} \frac{(n-1)!}{i!((n-1)-i)!} (F(y))^i (1 - F(y))^{(n-1)-i} \right) \end{aligned}$$

Now the two sums have the same expressions, all but one summand are cancelled out,

$$f_{(k)}(y) = n f(y) \left(\sum_{i=k-1}^{n-1} - \sum_{i=k}^{n-1} \right) \binom{n-1}{i} (F(y))^i (1 - F(y))^{(n-1)-i} = n f(y) \binom{n-1}{k-1} (F(y))^{k-1} (1 - F(y))^{n-k}.$$

This is the density formula for order statistics $Y_{(k)}, k = 1, \dots, n$.

Derivation of $\mathbb{E}[U_{(i)}] = \frac{i}{n+1}$

If $U_1, \dots, U_n$ are *i.i.d* $\sim \text{Uniform}([0, 1])$ and their order statistics are $U_{(1)} \leq \dots \leq U_{(n)}$, then $F(u) = \begin{cases} 0, & x < 0, \\ u, & u \in [0, 1], \\ 1, & u > 1, \end{cases}$

$f(u) = \begin{cases} 1, & u \in [0, 1] \\ 0, & \text{otherwise.} \end{cases}$ The density function of the order statistic $U_{(k)}$ has simpler form,

$$f_{(k)}(u) = n f(u) \binom{n-1}{k-1} (F(u))^{k-1} (1 - F(u))^{n-k} = n \binom{n-1}{k-1} u^{k-1} (1 - u)^{n-k}$$

Using the Gamma function $\Gamma(k) = \int_0^\infty x^{k-1} e^{-x} dx$ with $\Gamma(k) = (k-1)!$ for integer k , we can write the density as

$$f_{(k)}(u) = \frac{\Gamma(n+1)}{\Gamma(k)\Gamma(n-k+1)} u^{k-1} (1-u)^{(n-k+1)-1}, \quad u \in [0, 1]$$

This is the density function of Beta distribution with parameters k and $n-k+1$. That is,

$$U_{(k)} \sim \text{Beta}(k, n-k+1)$$

Calculate the mean of $\text{Beta}(k, n-k+1)$, we obtain

$$\mathbb{E}(U_{(k)}) = \frac{k}{n+1}$$

Derivation of $\mathbb{E}[\Phi(Z_{(i)})] = \frac{i}{n+1}$ in equation (1)

Apply Result (a) to standard normal random variable $Z \sim N(0, 1)$ with $\Phi(z) = \mathbb{P}(Z \leq z)$, we have

$$\Phi(Z) \sim \text{Uniform}([0, 1]), \quad i = 1, \dots, n$$

For $Z_1, \dots, Z_n$ *i.i.d.* $\sim N(0, 1)$, denote

$$U_i = \Phi(Z_i), \quad i = 1, \dots, n.$$

Then Result (a) and the *i.i.d.* property imply

$$U_1, \dots, U_n \text{ i.i.d. } \sim \text{Uniform}([0, 1])$$

By the strict monotonicity of Φ , we obtain the relation between the order statistics of Z_i 's and U_i 's,

$$U_{(k)} = \Phi(Z_{(k)}), \quad k = 1, \dots, n.$$

By Result (b) for order statistics of $U_i \sim \text{uniform}[0, 1], i = 1, \dots, n$, and $\mathbb{E}(U_{(k)}) = \frac{k}{n+1}$, we have

$$\mathbb{E}[\Phi(Z_{(k)})] = \mathbb{E}[U_{(k)}] = \frac{k}{n+1}, \quad k = 1, \dots, n.$$

□

If we approximate $Z_{(k)}$ by its approximated mean,

$$\mathbb{E}[Z_{(k)}] = \mathbb{E}[\Phi^{-1}(\Phi(Z_{(k)}))] \approx \Phi^{-1}(\mathbb{E}[\Phi(Z_{(k)})]),$$

we have the final approximation

$$\mathbb{E}[Z_{(k)}] \approx \Phi^{-1}\left(\frac{k}{n+1}\right), \quad k = 1, \dots, n.$$

Hence,

$$\left(\Phi^{(-1)}\left(\frac{i}{n+1}\right), x_{(i)} \right), \quad i = 1, \dots, n$$

are used in Normal Q-Q plot to check the *i.i.d.* normality assumption of a data set $\{x_1, \dots, x_n\}$.

Related reference: Rice 10.2.3, Casella & Berger 4.5.